ARL Spec: Mars Habitation Experiment

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1 Service overview

The Mars Habitation Experiment (MHE) is a 90-day isolation mission: a crew of five lives in a sealed, nine-room habitat that simulates a Mars base. Each crew member wears a vitals monitor, each room monitors its environment, and the crew fill in wellbeing surveys twice a day. Topic prefix: `arl/mhe`.

Entity kind	Ids	Notes
`room`	`quarters-1` to `quarters-5`, `common`, `lab`, `hydroponics`, `airlock`	The habitat's rooms (ids match the habitat plan `mhe.svg`). `quarters-n` is the private room of `crew-n`. Volumes: quarters 20 m³, common 90 m³, lab 60 m³, hydroponics 50 m³, airlock 12 m³.
`crew`	`crew-1` to `crew-5`	The crew members.
`service`	`global`	The habitat as a whole: mission clock, power, subsystems, alert summary.

1.1 General state and routines

- **Mission clock**: the mission runs for 90 days; `mission_day` counts from 1. The live deployment joins the mission part-way through (day 40), so earlier days are only in the ARL API backlog. After day 90 the surveys stop and `mission_state` becomes `complete`.
- **Reporting**: crew vitals every 60 s, room environment and the room-specific streams every 300 s (5 min).
- **Daily routine**: the crew follow a daily routine (local time) of sleep in their own quarters, meals and exercise in `common`, and work blocks in the `lab` or on `hydroponics` shifts. Crew `location` and room `crew_count` show where everyone is.
- **Surveys**: each crew member fills in a start-of-day survey in the morning and an end-of-day survey in the evening. A survey not done by 09:00 (start of day) or 23:30 (end of day) is missed.
- **EVAs**: about twice a week, two crew members go outside on a weekday afternoon for 2 to 3 hours. They enter the `airlock`, it depressurises (10 min), they go `outside`, return, and it repressurises (10 min). EVA start and end are published as events.
- **Rooms**: each room's temperature, humidity and ventilation follow its set points, and occupants add heat, humidity and CO_2. Sealing a room, disabling its environmental systems or a ventilation failure stops the air exchange, so CO_2 builds up in an occupied room. The airlock drops to Mars pressure (about 0.7 kPa) when depressurised.
- **Lighting**: each room has a daily lighting window and level (dim, 5 lx, outside the window). The hydroponics bay has strong grow lights.
- **Hydroponics**: a 400 L reservoir feeds the crops. It is drained by evaporation and the plants and refilled by the irrigation schedule. The crop goes through a 30-day cycle (germination, vegetative, flowering, harvest) with an estimated yield.
- **Load**: about 0.1 messages/s (5 vitals a minute, 13 room and habitat streams every 5 minutes).

1.2 Message conventions

Every service follows the same MQTT layout. `<entity>` is an entity id (or `global` for the whole service).

Topic	Retained	Content
`arl/mhe/$topology`	yes	Machine-readable list of the entities, streams (with field names, types, units and periods), state items, events and controls of the service. Re-published when it changes.
`arl/mhe/<entity>/telemetry/<stream>`	no	Periodic measurements.
`arl/mhe/<entity>/state/<item>`	yes	The current value of a state item, re-published when it changes.
`arl/mhe/events/<kind>`	no	One-off events, including `alert`.
`arl/mhe/cmd/<entity>/<control>`	no	Commands sent by the platform.
`arl/mhe/ack/<command_id>`	no	Acknowledgements of a command.

Telemetry and event payloads carry an envelope: `v` (schema version, integer, currently 1), `ts` (simulated time, Unix milliseconds), `seq` (per-topic sequence number, restarts with the service; gaps reveal lost messages) and `entity` (entity id). The measurement fields are added next to the envelope. A measurement that cannot be taken is `null`. State payloads are `{"v", "ts", "entity", "item", "value"}`.

Commands are JSON `{"command_id": <string>, "researcher": <researcher code>, "value": <value>}`. The researcher must be verified by the ARL API for the service. Acknowledgements carry `command_id`, `target`, `control`, `researcher`, `state` (`accepted`, `rejected`, `in_progress`, `completed`, `failed`), `progress` (0 to 1 for long-running controls), `reason` and `detail`. Rejection and failure reasons: `controls_disabled`, `malformed`, `unknown_control`, `unknown_target`, `unknown_researcher`, `researcher_defunct`, `researcher_not_permitted`, `api_unavailable`, `invalid_value`, `precondition_failed`, `busy` (the target is already running a long command), `fault`, `timeout`, `interrupted`.

Alerts are events on `arl/mhe/events/alert` with `alert_id`, `type` (the alert key below), `severity` (`info`, `warning`, `critical`), `state` (`raised` or `cleared`), `value`, `limit`, `unit`, `message` and `metadata`. An alert that escalates is raised again with the higher severity under the same `alert_id`. Limits come from the ARL API (`/v1/alert-limits/` + `mhe`).

Every service also publishes these:

Measurement	Key	Level	Type / unit	Description
Online	`online`	Global	boolean	Retained state. `true` while the service runs, `false` when it stops or loses its broker connection (last will).
Active alerts	`active_alerts`	Global	array of alert ids	Retained state. The ids of the alerts currently raised, in the order they were raised.

2 Transmitted data

2.1 Telemetry

2.1.1 Room environment (`arl/mhe/<room>/telemetry/environment`, every 300 s)

Measurement	Key	Level	Type / unit	Description
Temperature	`temperature_k`	Room	float, kelvin	Air temperature in the room.
Humidity	`humidity_pct`	Room	float, % relative	Relative humidity in the room.
Oxygen	`oxygen_ppm`	Room	integer, ppm	Oxygen concentration (about 209,000 ppm normally).
Carbon dioxide	`co2_ppm`	Room	integer, ppm	CO_2 concentration (about 420 ppm in an empty, ventilated room).
Pressure	`pressure_kpa`	Room	float, kPa	Air pressure (101.3 kPa; the airlock drops to about 0.7 kPa when depressurised).
Lighting	`lighting_lux`	Room	integer, lux	Current light level.
Ventilation rate	`ventilation_m3h`	Room	integer, m^3/h	Air exchanged with the habitat system. 0 when sealed, disabled or failed.
Crew count	`crew_count`	Room	integer, count	Number of crew in the room.

2.1.2 Hydroponics (`arl/mhe/hydroponics/telemetry/hydroponics`, every 300 s)

Measurement	Key	Level	Type / unit	Description
Water level	`water_level_l`	Room `hydroponics`	float, litres	Water in the 400 L nutrient reservoir.
Nutrient concentration	`nutrient_ec_mscm`	Room `hydroponics`	float, mS/cm	Electrical conductivity of the nutrient solution (about 1.2 to 2.4). Rises as water evaporates.
Water pH	`water_ph`	Room `hydroponics`	float, pH	Acidity of the nutrient solution (about 5.5 to 6.5).
Growth stage	`growth_stage`	Room `hydroponics`	enum	`germination`, `vegetative`, `flowering` or `harvest`.
Estimated yield	`estimated_yield_kg`	Room `hydroponics`	float, kg	Expected harvest of the current crop. Falls when water, nutrients or pH are out of range.

2.1.3 Airlock (`arl/mhe/airlock/telemetry/airlock`, every 300 s)

Measurement	Key	Level	Type / unit	Description
Airlock state	`airlock_state`	Room `airlock`	enum	`pressurised`, `cycling` or `depressurised`.
Inner door	`inner_door`	Room `airlock`	enum	`open` or `closed`. The door to the habitat.
Outer door	`outer_door`	Room `airlock`	enum	`open` or `closed`. The door to the outside.
EVA in progress	`eva_in_progress`	Room `airlock`	boolean	Whether crew are on an EVA.

2.1.4 Lab equipment (`arl/mhe/lab/telemetry/equipment`, every 300 s)

Measurement	Key	Level	Type / unit	Description
Equipment status	`equipment_status`	Room `lab`	enum	`ok`, `degraded` or `fault`.

2.1.5 Vitals (`arl/mhe/<crew>/telemetry/vitals`, every 60 s)

Measurement	Key	Level	Type / unit	Description
Heart rate	`heart_rate_bpm`	Crew	integer, beats/min	Heart rate from the wearable.
Body temperature	`body_temperature_k`	Crew	float, kelvin	Core body temperature (about 310 K).
Respiratory rate	`respiratory_rate_bpm`	Crew	integer, breaths/min	Breathing rate.
Blood oxygen	`spo2_pct`	Crew	float, %	Blood oxygen saturation. Drops in rooms with high CO_2.
Movement	`movement`	Crew	float, 0 to 1	Activity level: near 0 asleep, higher when working, exercising or on EVA.
Location	`location`	Crew	enum	Room id the crew member is in, or `outside`.
Heartbeat	`ts` (envelope)	Crew	timestamp, ms	Time of the latest vitals message. Vitals stop when the wearable is off or offline.

2.1.6 Habitat (`arl/mhe/global/telemetry/habitat`, every 300 s)

Measurement	Key	Level	Type / unit	Description
Mission day	`mission_day`	Global	integer, day	Day of the mission, 1 to 90.
Habitat power	`power_w`	Global	integer, watts	Electrical power used by the habitat.

2.2 State (retained)

Measurement	Key	Level	Type / unit	Description
Set points	`setpoints`	Room	object `{temperature_k` K, `humidity_pct` %, `ventilation_m3h` m^3/h}	Targets the room's environmental systems aim for.
Lighting schedule	`lighting_schedule`	Room	object `{start, end` `HH:MM`, `level_lux` lx}	Daily window when the lights are on, and their level.
Environmental systems	`environmental_systems`	Room	boolean	Whether heating, humidity control and ventilation are on in the room.
Sealed	`sealed`	Room	boolean	Whether the room is sealed from the habitat's air system.
Irrigation schedule	`irrigation_schedule`	Room `hydroponics`	object `{start, end`	Daily window when the reservoir is topped up, and the volume added per day.

Measurement	Key	Level	Type / unit	Description
			`HH:MM`, `volume_l` L}	
Subsystems	`subsystems`	Global	object of enums	Health of each habitat subsystem: `ventilation`, `heating`, `water`, `hydroponics`, `airlock`, `power_distribution`, `comms`, each `ok`, `degraded` or `fault`.
Mission state	`mission_state`	Global	enum	`running` or `complete`.

2.3 Events

Measurement	Key	Level	Type / unit	Description
Survey	`arl/mhe/events/survey`: `survey_type`, `mission_day`, `mood`, `stress`, `fatigue`, `sleep_quality`	Crew	enum, integer, integer scores 1 to 10	A completed wellbeing survey. `survey_type` is `start_of_day` or `end_of_day`. Higher is more for every score (high stress and fatigue are bad). `sleep_quality` is only in the start-of-day survey. A missed survey produces no event.
EVA	`arl/mhe/events/eva`: `event`, `crew`	Room `airlock`	enum, array of crew ids	An EVA `started` or `ended`, and the crew taking part.
Alert	`arl/mhe/events/alert`	Global / Room / Crew	alert object	Alert raised or cleared (see *Alerts*).

3 Controls

Control	Key	Level	Value (type / unit)	Description
Adjust Temperature	`adjust_temperature`	Room	float, kelvin, 285 to 300	Sets the room's temperature set point. The room approaches it over about half an hour.
Adjust Humidity	`adjust_humidity`	Room	float, %, 20 to 80	Sets the room's humidity set point.
Lighting Schedule	`lighting_schedule`	Room	object {`start`, `end` `HH:MM`, `level_lux` 0 to 1000 lx (hydroponics 0 to 30000)}	Sets the daily lighting window and level. The window may wrap past midnight. Lighting in quarters affects that crew member's sleep.
Ventilation	`ventilation`	Room	float, m^3/h, 0 to 400	Sets the room's ventilation set point.
Environmental Systems	`environmental_systems`	Room	boolean	Turns the room's heating, humidity control and ventilation on or off.
Seal Room	`seal_room`	Room	boolean	Seals or unseals the room. A sealed room exchanges no air, so CO_2 rises if it is occupied. The airlock cannot be unsealed while not pressurised.

Control	Key	Level	Value (type / unit)	Description
Crew Wellness Check	`crew_wellness_check`	Crew	boolean, must be `true`	Asks a crew member to check in. `in_progress` until they respond (minutes when awake, longer when asleep, after the EVA when outside), then `completed` with `detail` `ok` or `needs_support`.
Cycle Airlock	`cycle_airlock`	Room `airlock`	enum: `pressurise` or `depressurise`	Cycles the airlock (10 min `in_progress`). Rejected while a door is open, during an EVA, or when already in that state.
Irrigation Schedule	`irrigation_schedule`	Room `hydroponics`	object {`start`, `end` `HH:MM`, `volume_l` 0 to 120 L}	Sets the daily window and volume of water added to the reservoir.

4 Alerts

Alert	Key	Level	Severity, limit key (default, unit)	Description
Crew Health	`crew_health`	Crew	warning, critical; `mhe.crew.hr_min` (40 bpm), `mhe.crew.hr_max` (150 bpm), `mhe.crew.temp_max_k` (311.15 K), `mhe.crew.resp_max` (30 / min), `mhe.crew.spo2_min` (92 %)	A vital sign outside its limit for 3 min; critical when two or more are. `metadata.vital`, `metadata.count` and `metadata.location`. Clears when all are inside the limits for 5 min.
Crew Status	`crew_status`	Crew	warning; no limit	No vitals from the wearable for 5 min (`metadata.reason` `wearable_offline`). Clears on the next vitals message.
Survey Missed	`survey_missed`	Crew	info; no limit	A survey was not completed by its deadline (09:00 or 23:30). Clears on the next completed survey.
Survey Wellbeing	`survey_wellbeing`	Crew	warning; `mhe.survey.min_mood` (4, score), `mhe.survey.max_stress` (7, score), `mhe.survey.max_decline` (2, score)	Judged daily at 23:30: the 3-day mean mood is below the limit, the 3-day mean stress is above it, or mood has dropped by more than the decline limit compared with a week earlier. Clears when all are back inside.
Room Condition	`room_condition`	Room	warning; `mhe.room.temp_min_k` (290 K), `mhe.room.temp_max_k` (298 K), `mhe.room.humidity_min` (25 %), `mhe.room.humidity_max` (75 %), `mhe.room.humidity_max_hydroponics` (85 %)	Temperature or humidity outside the limits for 15 min (hydroponics uses its own upper humidity limit). Clears 0.5 K and 3 % inside the limits.

Alert	Key	Level	Severity, limit key (default, unit)	Description
Air Concentration	`air_concentration`	Room	warning, critical; `mhe.air.o2_min_ppm` (195000 ppm), `mhe.air.co2_warn_ppm` (1500 ppm), `mhe.air.co2_crit_ppm` (2500 ppm)	O_2 below or CO_2 above the limits for 10 min; critical above the second CO_2 level. `metadata.crew_count` and `metadata.sealed`. Clears inside the limits for 5 min.
Habitat Fault	`habitat_fault`	Global	critical; no limit	A habitat subsystem is in `fault` (`metadata.subsystems` lists them). Clears when every subsystem is `ok` or `degraded`.

The airlock is not judged for Room Condition or Air Concentration while it is cycling or depressurised.

5 Configuration

The service reads a TOML file (`config/` + `mhe.toml` in the bundle, `--config` or `ARL_CONFIG` for a binary). Environment variables override the file, and command-line options override both (see *Running the ARL*). Defaults are the values in the provided file.

5.0.1 Common keys

Key	Type / unit	Default	Description
`seed`	integer	`1`	Random seed. The same seed, start time, scenario and commands give the same data. Env `ARL_SEED`.
`speedup_multiplier`	float, ×	`1.0`	Simulation speed relative to real time, up to the service's `max_speedup`. Env `ARL_SPEEDUP`.
`control_enabled`	boolean	`true`	Whether the service accepts commands. When `false` every command is rejected (`controls_disabled`). Env `ARL_CONTROL_ENABLED`.
`start_time`	`"now"` or RFC 3339	`"now"`	Simulated time at start-up. Env `ARL_START_TIME`.
`scenario`	string, path	`""`	Scenario file to load, empty for none. Env `ARL_SCENARIO` (compose file).
`mqtt.address`	string	`"mosquitto"`	MQTT broker host. Env `MQTT_BROKER_ADDRESS`.
`mqtt.port`	integer	`1883`	MQTT broker port. Env `MQTT_BROKER_PORT`.
`mqtt.user`	string	`""`	Broker username, empty for anonymous. Env `MQTT_BROKER_USER`.
`mqtt.password`	string	`""`	Broker password. Env `MQTT_BROKER_PASSWORD`.
`mqtt.queue_capacity`	integer, messages	`200000`	Outgoing messages the service buffers while the broker is slow or unreachable.
`api.address`	string	`"api"`	ARL API host. Env `ARL_API_ADDRESS`.
`api.port`	integer	`8080`	ARL API port. Env `ARL_API_PORT`.
`api.require_researcher`	boolean	`true`	Whether commands must carry a researcher code the API accepts for this service.

Key	Type / unit	Default	Description
`api.researcher_cache_ttl_s`	integer, s	`60`	How long a researcher check is remembered.
`api.alert_limits_refresh_s`	integer, s	`300`	How often the alert limits are re-read from the API.
`health.port`	integer	`9100`	Port of the service's `/healthz` endpoint (used by `healthcheck`). Env `HEALTH_PORT`.

5.0.2 MHE keys (`[mhe]`)

Key	Type / unit	Default	Description
`crew_size`	integer, 1 to 5	`5`	Number of crew members (`crew-1` upwards), one per quarters.
`mission_days`	integer, days	`90`	Length of the mission.
`start_day`	integer, day	`1`	Mission day the service starts on. Set it higher to start mid-mission.
`crew_frequency_hz`	float, Hz	`0.0167`	Reporting rate of the vitals stream (whole minutes; 0.0167 Hz is every minute).
`room_frequency_hz`	float, Hz	`0.0033`	Reporting rate of the room and habitat streams (0.0033 Hz is every 5 minutes).
`eva_per_week`	integer	`2`	EVAs per week.
`max_speedup`	float, ×	`2000`	Highest `speedup_multiplier` the service accepts.

6 Scenario events

Events with `service = "mhe"` (format in *Running the ARL*).

Fault	Entity	Params (default)	Effect
`ventilation_fault`	Room	none	The room's ventilation stops (ventilation subsystem degraded or fault; CO_2 builds up if occupied).
`heater_fault`	Room	none	The room's heating fails and its temperature drifts down (Room Condition).
`sensor_drift`	Room	`quantity` (`temperature` or `humidity`), `rate` (0.2 per hour)	The room's temperature or humidity reading drifts away from the true value.
`hydroponics_pump_fault`	none	none	Irrigation stops (hydroponics subsystem degraded).
`airlock_door_fault`	none	none	The airlock door reports open when it should not (airlock subsystem). Lasts until the airlock is cycled to `pressurise` or a `repair` on `airlock`.
`subsystem`	none	`subsystem` (name), `status` (`ok`, `degraded`, `fault`)	Forces a habitat subsystem's status for `duration` (`fault` raises Habitat Fault).
`lab_equipment`	none	`status` (`ok`, `degraded`, `fault`)	Sets the lab equipment status.

Fault	Entity	Params (default)	Effect
`wearable_offline`	Crew	none	The crew member's wearable stops sending vitals (Crew Status).
`crew_illness`	Crew	`fever_k` (1.5 K)	The crew member falls ill: fever and raised heart rate (may raise Crew Health).
`skip_survey`	Crew	none	The crew member skips their next survey (Survey Missed at the deadline).
`wellbeing_drop`	Crew	`delta` (0.3)	The crew member's surveys score worse for `duration` (may raise Survey Wellbeing).
`repair`	Room or crew	none	Ends the faults on the room (including the hydroponics pump and the airlock door for those rooms) or on the crew member.

AI Declaration: Claude Code (Opus 5.5) - generated from the arl-sim source, reviewed